Hariton Pitsik

ML ENGINEER · AI RESEARCHER · DATA SCIENTIST

Saratov, Russia

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Education

Saratov State University Saratov, Russia

BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY September 2023 - June 2027

Skills

Programming languages	Python, Rust, SQL
Spoken languages	Russian (native), English (B1-B2)
Technologies	Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL
Concepts	Linear Algebra, Mathematical Analysis, Probability Theory & Statistics, Computer Science, Classical ML Algorithms (Supervised & Unsupervised), Deep Learning, Natural Language Processing (NLP), Large Language Models (LLM)

Some achievements

Teaching & Social

- Participant of Development Students Club (DSC)
- I believe education is important and should be shared. I teach machine learning as part of a DSC community.
  - Lecture recordings on NumPy (YouTube) and Pandas (YouTube).
  - Machine Learning Club playlist: (YouTube)

Competitions

- Participant in the HSE & X5 Growth Gradient contest, focused on time series forecasting (Link).
- Participant in the FSP hackathon as an analyst of existing hashing methods and potential sources of entropy.

Projects

Local RAG System — github.com/haritonn/pdf_rag

- Built a local RAG system with a switchable LLM backend;
- Implemented hybrid retrieval, document chunking, and automated relevance evaluation;
- Integrated the Qdrant vector database and built a Streamlit interface.

Transformers from Scratch — github.com/haritonn/transformers-scratch

- Implemented the full encoder-decoder architecture in PyTorch;
- Built a modular training pipeline for seq2seq tasks;
- Designed a fully customizable pipeline for experiments.

Caption Generator — github.com/haritonn/caption_gen

- Built an image captioning train pipeline with a ResNet backbone and an attention-based LSTM;
- Implemented the project entirely in PyTorch without pre-built components;
- Implemented beam search and greedy decoding;

- Built an end-to-end training workflow.

ResNet with Attention
github.com/haritonn/resnet_attention

- Implemented baseline ResNet and attention-augmented variant in PyTorch;
- Compared convergence speed and classification accuracy;
- Tracked experiments with ClearML.

Coursework

2nd Year: Generative Computer Vision
github.com/haritonn/coursework2

Compared diffusion models, GANs, and large multimodal model (LMM)-based approaches for virtual try-on. Evaluated visual quality and inference time on a custom dataset.

3rd Year: Analysis of Geometric Properties of Embeddings

Ongoing coursework focused on the analysis of embedding spaces.